

1-C:

The correct answer is C) It is not possible to produce the level of output at a lower unit cost.

Productive efficiency refers to the optimal use of resources to **produce a given level of output at the lowest possible cost**. In other words, it means that firms are producing at the minimum point of their average total cost curve.

Option C best defines productive efficiency because it indicates that the current level of output is being produced at the lowest possible unit cost, and any changes in production would increase the cost per unit.

A) This statement relates to profit maximization, not productive efficiency.

B) This statement describes Pareto efficiency.

D) This statement describes the production possibility frontier (PPF), which represents the various combinations of output of two goods that an economy can produce given its resources and technology.

2-C:

- Overconsumption of demerit goods (Yes): Sugary drinks are considered demerit goods, which are goods that are harmful to consumers. Imposing a tax can reduce consumption and discourage harmful behavior.
- Negative externalities of consumption (Yes):
- **The question says that drinking too many sugary drinks is bad for health, which represents a problem arising from consumption of sugary drinks.** The consumption of sugary drinks can lead to negative externalities, such as increased healthcare costs and social burdens, which are not reflected in the market price. A tax can internalize these externalities and account for the social costs.
- Negative externalities of production (No): The production of sugary drinks may have negative externalities of production, but a **tax on consumption** would not directly address these issues.

3-D:

$0.5 + 1.8 \rightarrow 2.3$

↑
Social benefit
= PB + EB

		\$ billion
private costs		1.0
social costs		1.7
private benefits		0.5
external benefits		1.8

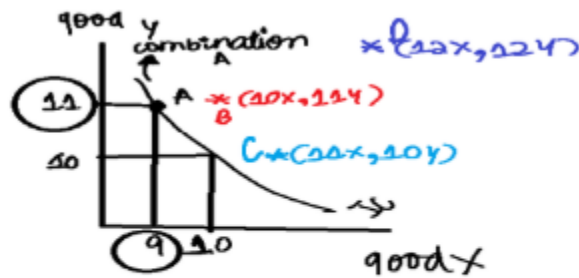
What should the government decide and why?

	decision	reason
A	do not proceed	total costs exceed external benefits
B	do not proceed	total costs exceed total benefits
C	proceed	private costs exceed private benefits
D	proceed	social benefits exceed social costs

Handwritten notes and calculations:

- $0.5 + 1.8 \rightarrow 2.3$ (Social benefit)
- $1.0 + 0.7 = 1.7$ (Social costs)
- $2.3 - 1.7 = 0.6$ (Net social benefit)
- $SB > SC$ (Social benefit > Social cost)
- $(2.3) > (1.7)$
- 0.6 (Net social benefit)
- socially worthwhile**
- proceed**

4-A:



It can be seen in the diagram above that only combination **A** lies on the same indifference curve, whereas, all other combinations (**B, C and D**) lie on the other indifference curves

5-B:

Since the budget line has shifted rightwards parallelly, this represents a “rise in real income”

A - In-order to identify if a good is a giffen good, we need to observe the income effect and the substitution effect of changes in the price of the good. This won't be identified if changes in incomes have occurred.

B – Here, after a rise in real income, the quantity of good X is shown to contract, which means that the good is an inferior good, since there is a negative relationship between income and demand of inferior goods. This is why, this becomes correct.

C – It is good Y, which is a “normal good” because a rise in real incomes has resulted in an extension in qty demanded of good Y.

D – A public good is a good which exhibits the characteristics of non-excludability and non-rivalry, and also non-reject-ability, and a “free-rider problem” is also associated with it. This cannot be deduced from the information available.

6-C:

The correct formulae for calculating average cost (AC) and average revenue (AR) are:

- Average Cost (AC) = Total Cost (TC) / Quantity (Q) = TC/Q
- Average Revenue (AR) = Total Revenue (TR) / Quantity (Q) = TR/Q or $(P*Q)/Q$

This explains why key C will be correct.

7-A:

In both, perfect competition and monopolistic competition, firms make normal profits in the long-run. In both monopolistic competition and perfect competition, firms will reach a point of equilibrium in the long run where:

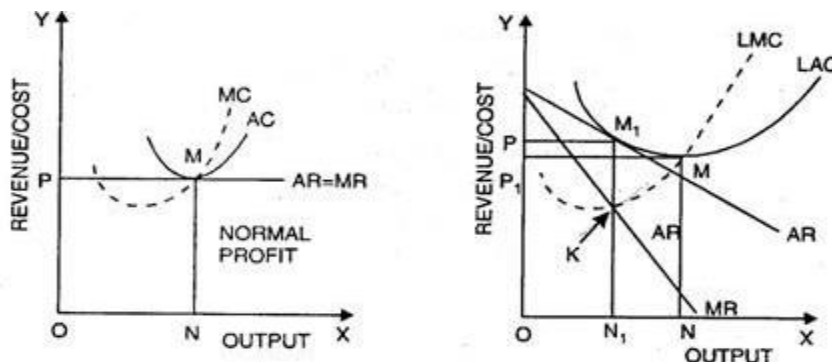
- Average Revenue (AR) = Average Cost (AC)

This means that the total revenue earned by the firm is equal to the total cost of production, and the firm is earning zero economic profit.

B) It is only in perfect competition that marginal revenue i.e. the revenue earned upon selling an additional unit is equal to Average revenue i.e. price. This is because, price remains the same under conditions of perfect competition. Since monopolistic competition is a part of imperfect competition, Average revenue will not be equal to Marginal revenue.

C) When Average cost is equal to Marginal cost, this represents that Average cost has been minimized, or in other words, productive efficiency has been achieved. This is only achieved under perfect competition in the long-run.

D) Average cost is not equal to marginal revenue under conditions of monopolistic competition. This is why, key D, is incorrect.



8-C:

- A- This is incorrect because Average fixed cost can never be zero.
- B- When Average revenue is equal to Average cost, this represents that only normal profits are being made, and so, the objective of profit maximization is not being achieved.
- C- The profit maximizing level of output is where Marginal cost (MC) = Marginal revenue (MR). And revenue is maximized when $MR=0$
 Let's substitute the revenue maximizing equation in the profit maximizing equation.
 $MC=MR(0)$
 This represents that MC will also need to be zero for revenue maximization and profit maximization to be achieved at the same time
 $MC(0) = MR(0)$.
 This is why, key C, will be correct.

D – When $MR > MC$, this represents that profits are not maximized, and more needs to be produced until the level of output where $MC=MR$.

9-A:

The correct answer is A) It will shift to the left at any given price.

When firms leave an industry, the total production capacity of the industry decreases, leading to a **decrease in the quantity supplied at each price level**. As a result, the supply curve shifts to the left.

B) A shift to the right indicates an increase in the quantity supplied at each price level, which occurs when new firms enter the industry.

C and D) Downward and upward movement along the supply curve represent the effect on quantity supplied arising as a result of price changes.

10-D:

If a firm has limited access to financial capital, it may not be able to expand its operations or invest in new technologies, making it difficult to grow and remain competitive. This can be a reason for a firm to remain small. This explains why key D will be correct.

A) A high minimum efficient scale (MES) means that firms need to produce at a large scale to be efficient, which can lead to economies of scale and encourage firms to grow.

B) X-inefficiency refers to the phenomenon where firms are not operating at their maximum efficiency due to lack of competitive pressures. This mainly occurs in a monopoly market structure.

C) Significant economies of scale encourage firms to grow and expand their operations to take advantage of cost savings and increased efficiency.

11-D:

The principal-agent problem refers to the situation where the owners of a firm (principals) have different goals and interests than the managers (agents) who are responsible for running the firm. This can lead to conflicts, as managers may prioritize their own interests over the interests of the owners. **Mostly managers prefer the objective of sales revenue maximization because their salaries and bonuses are linked to revenues and not profits, contrary to the shareholders' preference of profit maximization.**

A) The disincentive effect of high tax rates on entrepreneurs is a problem of taxation, not a principal-agent problem.

B) The existence of trade unions is a way for workers to negotiate with management, but it's not a principal-agent problem.

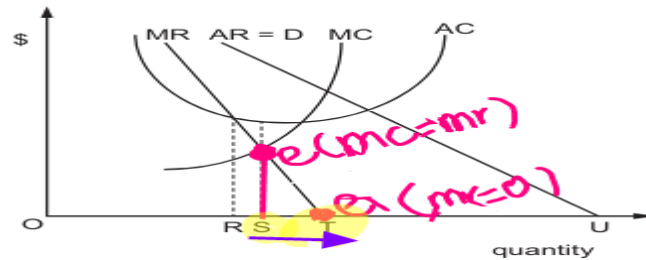
C) The lack of consumer knowledge about product quality is an example of asymmetric information, not a principal-agent problem.

12-C:

When the objective was profit maximization, the firm's profit maximizing level of output was S, because that was when **marginal cost was equal to marginal revenue.**

Similarly, when the objective changed to revenue maximization, the level of output was T, because that was when **marginal revenue was equal to zero.**

This represents that the firm's output will increase from S to T after this change.



13-C:

C) Governments increase tax rates on high earners to reduce income inequality and redistribute incomes from the rich to the poor, **prioritizing equity over efficiency**. This approach aims to make the tax system more progressive, where those with higher incomes contribute a greater share of their income towards taxes.

A) If the government expects total tax revenue to be unaffected, it would not change tax rates.

B) Had the government been concerned about the possibility of emigration by some high earning individuals, it would not have increased the income tax rate.

D) Switching emphasis from direct to indirect taxation would involve increasing taxes like VAT or sales tax, not income tax.

14-B:

The correct answer is B) poverty trap.

When benefits are reduced as income rises, and income tax is introduced, it can create a **poverty trap**. This means that individuals may be discouraged from working more or earning higher incomes because they would lose benefits and face higher taxes, effectively trapping them in poverty.

15-D:

A – When this protectionist measure encourages domestic steel producers to achieve economies of scale, this will result in **an increase in productive efficiency**.

B - Preventing dumping of low-grade steel can protect the interests of domestic consumers, which can contribute to an increase in consumer satisfaction that aligns with allocative efficiency.

C – When it protects jobs of highly productive steel workers, this will prevent a loss of productive efficiency from occurring.

D - Imposing a minimum price on imported steel can reduce competition in the domestic market and these lack of competitive pressures may lead to X-inefficiency, which may result in firms voluntarily choosing not to operate on the lowest point of the ATC curve and choosing to forego productive efficiency and when this results in higher costs of building programmes, this will represent inefficiency.

16-C:

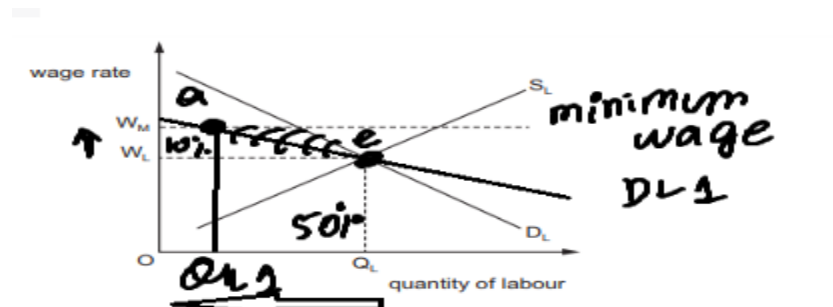


This explains why, key C, will be correct.

17-B:

A – If the demand for labor rises, this will be of a benefit to workers, because in this case, a rise in wage rate will be accompanied by a rise in employment.

B - If the price elasticity of demand for workers was elastic “shown through $DL1$ ” (or in other words, high), then in that case, a rise in wages arising from minimum wages will result in a contraction in quantity demanded for labor by a relatively higher percentage, resulting in a higher proportionate increase in unemployment of existing workers. This is why, key B, becomes correct.



C – If the elasticity of supply was very high, then this would have led to extension in qty supplied by a relatively higher percentage resulting in unemployment of new entrants. Unemployment of existing workers will be based on the demand for labor.

D - If the demand for labor rises as a result of a rise in productivity of workers, this will be of a benefit to workers, because in this case, a rise in wage rate will be accompanied by a rise in employment.

18-B:

A) Economic growth indeed creates costs (e.g., environmental degradation, increased inequality) alongside benefits (e.g., higher incomes, improved living standards).

B) Long-term economic growth is typically measured by calculating the average annual percentage increase in GDP over a period of years . Measuring growth over a single year (as stated in Key B) does not capture long-term trends and can be misleading due to short-term fluctuations, such as business cycles, seasonal variations, or one-time events .**This is why, this becomes the correct answer.**

C) Short-term economic growth or **actual economic growth** can be represented by a shift from a point within a production possibility curve (actual) to a point on the curve (potential), indicating increased efficiency and resource utilization.

D) Sustainable economic growth is indeed defined as growth that does not compromise the ability of future generations to meet their own needs, by depleting resources or causing irreparable environmental harm.

19-C:

Real GDP per head is an imperfect measure of welfare in developed economies because it only captures a narrow aspect of economic activity, namely the production of goods and services after having adjusted it for inflation and the size of the population. It does not account for:

- Income inequality and distribution
- Non-market activities (e.g., household work, volunteering)
- Environmental degradation and resource depletion
- Leisure time and work-life balance
- Health and education outcomes

A) While government manipulation and misuse of data can be a concern, it is not the reason why real GDP per head (specifically) is considered an imperfect measure of welfare.

B) Real GDP per head is adjusted for inflation, so this is not a concern.

D) The shadow economy (unreported economic activity) not being recorded comes under the point that the scope of the real gdp per head data is restricted, which makes option C, the correct choice.

20-B:

B - Progressive direct taxation means that a greater share of the incomes of higher income higher income individuals are taken up in tax because the average rate of tax increases upon a rise in income. Since this reduces the after-tax income of the higher income groups, whilst making funds available for the government to redistribute to the poor through a system of means tested benefits, this will reduce income inequality.

Decreased indirect taxation (such as sales taxes or VAT) benefits lower-income individuals, since they are regressive in nature, which means that they take up a greater proportion of the incomes of low-income individuals.

A) Additional government regulation of product and factor markets may not necessarily reduce income inequality.

C) Increased trade union power and reduced labor market flexibility may lead to **higher wages for unionized workers** but may not address overall income inequality.

D) Reduced welfare benefits would likely increase income inequality, as vulnerable populations would be negatively affected.

21-B:

The formula for labor force is:

Employed+ unemployed

$$- 15 + 0.5 = 15.5$$

This is why key B is correct.

22-D:

Nationally negotiated wage rates can lead to higher labor costs in regions with high unemployment, making it harder for businesses to operate and expand. By ending these national wage rates, businesses in the region can negotiate lower wages **in light of the surplus of the labor available**, making it more feasible to invest and expand in the area, creating jobs and stimulating industrial expansion.

A) A rationing system for building development would limit industrial expansion, not encourage it.

B) An increase in taxes on investment would discourage industrial expansion, not encourage it.

C) A requirement that only local labor may be employed could lead to labor shortages, hindering industrial expansion.

23-C:

A) Building larger fishing boats may lead to overfishing, depleting marine resources.

B) Clearing rainforests for agriculture would harm the environment, which represents unsustainable growth.

C) Given the island economy's limited agricultural land, training farmers and agricultural workers to increase crop yields is a sustainable way to improve agricultural productivity, without expanding into rainforests or relying on finite resources. This explains why key C is correct.

D) Using domestic mineral resources to rapidly increase manufacturing output may lead to resource depletion and environmental degradation, rather than sustainable growth.

24-C:

The correct answer is C) 20,000.

To calculate national income, we need to add up all the income received by factors of production:

- Wages: 8,000
- Salaries: 7,000
- Rent: 3,000
- Interest: 2,000

Total factor income = 20,000

Note that unemployment benefits and government pensions are transfer payments, not factor incomes, so they are not included in the calculation of national income.

25-D:

The formula to find an increase in Real national income will be:

Increase in Aggregate expenditure * Multiplier = Increase in Real national income.

Multiplier = $1 / (1 - MPC)$ -> since we have a two sector economy, we are interested in the formula of two sector multiplier.

$$\text{Multiplier} = 1 / (1 - 0.6) = 1 / 0.4 = 2.5$$

The increase in national income will be:

$$\begin{aligned} \text{Increase in national income} &= \text{Increase in investment} \times \text{Multiplier} \\ &= \$100 \text{ million} \times 2.5 \\ &= \$250 \text{ million} \end{aligned}$$

26-A:

The correct answer is A) no yes

According to the liquidity preference theory, idle balances (also known as speculative balances) are responsive to a change in the interest rate, which represents that **speculative demand for money is interest elastic.**

On the other hand, **active balances (also known as transactionary or precautionary balances)** are not responsive to changes in interest rates, as they are held for everyday transactions or are kept as a precaution against emergencies. They are not influenced by interest rate fluctuations. Therefore, the transactionary or precautionary demand for money is relatively interest inelastic, because it is influenced by the level of personal income.

27-C:

The credit multiplier is calculated as:

$$\text{Credit multiplier} = 1 / \text{reserve ratio}$$

Initially, the Cash Deposit Ratio is 5%

So, the initial Credit Multiplier is:

$$CM = 1 / 0.05 = 20$$

When the Cash Deposit Ratio increases to 10% = 0.10, the new Credit Multiplier becomes:

$$CM = 1 / 0.10 = 10$$

To find the percentage change in the Credit Multiplier, we can use the formula:

Percentage change = $((\text{New value} - \text{Old value}) / \text{Old value}) \times 100$

Percentage change in CM = $((10 - 20) / 20) \times 100 = -50\%$

So, the Credit Multiplier falls by 50% when the Cash Deposit Ratio increases from 5% to 10%.

28-C:

While removing health and safety regulations might lead to an increase in economic growth in the short term (as companies may reduce costs and so Aggregate supply may be increased), it is likely to restrict economic development in the long term. This is because neglecting health and safety can lead to workplace accidents, illnesses, and environmental degradation, ultimately harming the workforce and the environment.

A) Increasing healthcare facilities cannot only improve productivity and increase economic growth but may also boost economic development.

B) Raising the school leaving age can lead to a more educated workforce, driving economic development.

D) Promoting export-led growth can stimulate economic growth and development by encouraging industries that compete globally.

29-A:

A) In the short run, increasing spending on education and health can lead to an increase in aggregate demand, and if the economy is already operating at or near full capacity, this will lead to a situation where Aggregate demand outpaces Aggregate supply, and these shortages can contribute to an increase in general price level, namely "demand pull inflation"

B) Increasing the basic rate of income tax could reduce disposable income, decrease consumption, and subsequently bring about a fall in Aggregate demand, which can actually help reduce inflationary pressures.

C) Limiting wage increases in the public sector can help control labor costs and inflation.

D) Preventing mergers of large companies can promote competition, which can help keep prices in check.

30-D:

A – A decrease in the rate of interest will increase investment after having decreased the costs of borrowing, which will increase Aggregate demand.

A resultant rise in incomes will result in a higher import spending, which can reduce the bop surplus, but there will still be a conflict with price stability as when Aggregate demand outpaces Aggregate supply, there will be shortages that will result in demand-pull inflation.

B - A devalued currency would make exports cheaper and more competitive, while making imports appear more expensive. This will result in an increase in export revenues and a decrease in import expenditures in the long-run when the Marshall-Lerner condition is held, which will lead to an increase in the trade surplus. Since this represents net injections into the circular flow, there will be an increase in Aggregate demand and when Aggregate demand outpaces Aggregate supply, there will be shortages that will result in demand-pull inflation.

C – Although an increase in direct taxes will reduce disposable incomes and subsequently decrease consumption, there will be a fall in Aggregate demand, which will dampen inflationary pressures, and it will also bring about a decrease in demand for imported goods, which will lead to an increase in the trade surplus.

D – Removal of tariffs on imports will lead to an increase in import spending, and a subsequent increase in import expenditures will lead to the trade surplus being reduced. Additionally, when because of this, domestic consumption and Aggregate demand falls, inflationary pressures will also be dampened.

